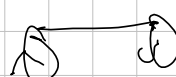
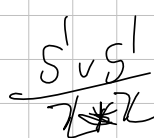
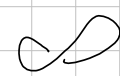
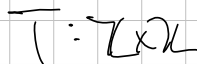
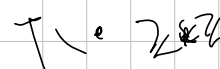
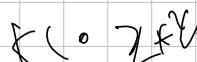


$(\mathbb{Z} \times \mathbb{Z})$
 $aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$






[illegible]

R/P^2 : 0-cell: $\{v\}$
 1-cell: $\{w\}$
 2-cell: $\{w\}$

R/P^2 : $\mathbb{Z}_2$ $\langle a \mid a^2 \rangle$
 $a^2 = 1$

R/P^2 : 0-cell: $\{v\}$
 1-cell: $\{w\}$
 2-cell: $\{w\}$

$\pi_1(R/P^2) = \mathbb{Z}_2$